CSET485 – AI and Society

Assignment 1: Stress-Testing a Predictive Model

Roll No. 011 | **Dataset** COMPAS Recidivism Risk Score (ProPublica, compas-scores-two-years.csv) | **Random seed** 11

This report is for the COMPAS Recidivism Risk Score dataset. My roll number ends in 011, so its last digit (1) assigns this dataset and random_state is 11. Every number below comes from my notebook's printed output, from the CSV files it saves in *compas_assignment_outputs*, or from simple descriptive checks on the same data. The few values I worked out by hand from these are labelled as derived. The binary target is *two_year_recid* (1 = reoffended within two years), which stands in for the *is_recidivist* column named in the assignment. Here the flagged group is the 'high risk' group, meaning defendants predicted to reoffend, so being flagged is the bad outcome for the person (the opposite of being approved for a loan).

Three results need a little extra care, so I want to say them up front. First, the Simpson's search found **no sign reversal** in the key predictor priors_count; the clearest subgroup difference is in sex_Male ($z = 2.58$, $p = 0.0098$). Second, dropping race moved no predictor's coefficient by more than 3.01%, so the omitted-variable test only reassures me about those four predictors. Third, at threshold 0.05 nearly everyone is flagged, so DP = EO = 1.0 there tells us almost nothing.

1. Data Preprocessing

• Dataset and target

The ProPublica file has 7,214 defendants and 53 columns. I used seven of them: decile_score (Model A target), two_year_recid (Model B target), age, priors_count, sex, race and c_charge_degree. The target is 1 for 3,251 defendants (45.07%) and 0 for 3,963. decile_score runs from 1 to 10 (mean 4.51).

• Missing values

There are no missing values in the seven columns used (null count 0 for each), and the number of rows used (7,214) equals the raw row count, so no rows were removed. The pipeline still contains a safeguard imputer (median for numeric, most-frequent for categorical), which changed nothing here.

• Categorical encoding

sex (Male/Female), race (six levels) and c_charge_degree (F/M) were one-hot encoded with the first category dropped, so the baselines are Female, African-American and Felony. This gives nine model features: age, priors_count, sex_Male, five race dummies (Asian, Caucasian, Hispanic, Native American, Other) and c_charge_degree_M. Race counts are African-American 3,696 (51.2%), Caucasian 2,454 (34.0%), Hispanic 637, Other 377, Asian 32 and Native American 18, so coefficients for the last two groups are very unreliable.

• Outliers

I did not remove or winsorise any row. Ages go up to 96 and priors_count up to 38 (skewness 2.37; 736 defendants, 10.2%, have 10 or more priors). These are real defendants, and dropping them would hide exactly the cases that the adversarial test in Section 3.3 looks at, so I kept them on purpose. Model A's predictions do fall outside the 1 to 10 scale (−2.47 to 12.21), which is discussed in Part 3, Question 1.

• Scaling and split

The split came first. With random_state = 11, a 70/30 split stratified on the target gave 5,049 training and 2,165 test rows (45.06% and 45.08% reoffenders). age and priors_count were standardised by a StandardScaler inside the pipeline and fitted on the training rows only, so nothing from the test set leaks into the models. The dummies stay 0/1.

Note for comparing sections: the statsmodels audits in Sections 3.1 and 3.2 standardise age and priors_count within the sample being fitted (all 7,214 rows for the full and reduced models, each race group on its own for the subgroup models), so those continuous coefficients are per standard deviation of that sample. Binary predictors such as sex_Male are unaffected. These audit models use all 7,214 rows, not just the training set.

2. Part 1: Build and compare

Model A is OLS linear regression (sklearn LinearRegression) predicting decile_score; its test predictions are thresholded at their own median, **4.6267**. **Model B** is logistic regression (max_iter = 3000, default L2 penalty) predicting two_year_recid; its probabilities are thresholded at 0.5. Performance of both rules on the 2,165 test defendants:

Table 1. Test-set performance of the two classification rules (flagged = high risk).

Rule	Flagged high risk	Accuracy	Precision	Recall	ROC-AUC
Model A: OLS, median rule (≥ 4.6267)	1,083 (50.02%)	0.6420	0.5928	0.6578	not applicable
Model B: logistic, P ≥ 0.5	792 (36.58%)	0.6628	0.6553	0.5318	0.7095

Model B accuracy and ROC-AUC are printed by the notebook; the other entries are derived from the saved part1_test_predictions.csv.

Agreement between the two thresholding approaches (rows = linear median rule, columns = logistic 0.5 rule):

Table 2. Agreement between the median-threshold OLS rule and the 0.5-threshold logistic rule.

	Logistic = low	Logistic = high	Total
Linear = low (below median)	1,066	16	1,082
Linear = high (≥ median)	307	776	1,083
Total	1,373	792	2,165

Mismatched cases: 323 of 2,165 (14.92%). Matching cases: 1,842 (85.08%). Of the 323 mismatches, 307 (95.0%) are defendants the linear rule flags but logistic regression does not; only 16 go in the other direction.

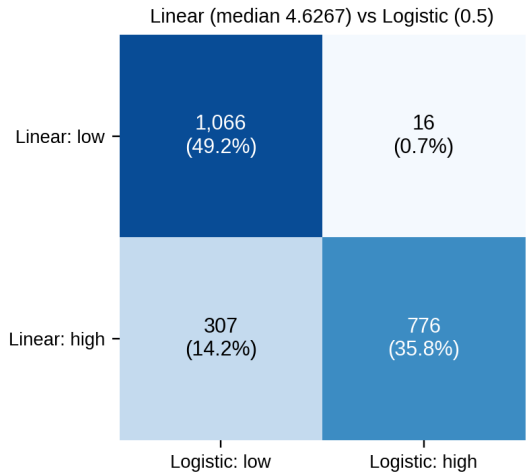


Figure 1. Confusion matrix of the two classification rules.

Why they differ. See Part 3, Question 1: the linear median (4.6267 on a 1–10 scale, about 0.40 once rescaled to 0–1) sits below the logistic 0.5 cut-off, and a median rule forces a 50/50 split while only 45.08% of test defendants reoffended.

3. Part 2: The trap

• 3.1 Simpson's Paradox hunt

I split on race and compared African-American defendants (Group A, 3,696) and Caucasian defendants (Group B, 2,454) with the full dataset (7,214), fitting a logistic regression with all five features in each case. The key predictor is priors_count.

Table 3. priors_count coefficient (standardised, per SD of the fitted sample) by race group.

Model	n	priors_count coef.	95% CI	p-value	Odds ratio per SD
Full dataset	7,214	0.7357	[0.6698, 0.8016]	3.5e-106	2.087
African-American	3,696	0.7905	[0.6973, 0.8836]	4.1e-62	2.204
Caucasian	2,454	0.6816	[0.5743, 0.7889]	1.4e-35	1.977

The coefficient **does not change sign**: it is positive and strongly significant in all three models. It is somewhat larger for African-American defendants, but the intervals overlap, and a Wald-type test on the difference (0.1089; standard errors recovered from the two intervals) gives z = 1.50, p = 0.133, which is not significant. Since priors_count does not reverse, I checked which predictor's effect differs most between the two groups, using the subgroup coefficient table my notebook saves:

Table 4. Other predictors across the three models, with the Group A minus Group B difference (Wald z from CI-derived standard errors).

Predictor	Full dataset (95% CI)	African-American (95% CI)	Caucasian (95% CI)	A − B difference
priors_count	0.7357 [0.6698, 0.8016]	0.7905 [0.6973, 0.8836]	0.6816 [0.5743, 0.7889]	+0.1089; z = 1.50, p = 0.133
sex_Male	0.3126 [0.1836, 0.4416]	0.4852 [0.3014, 0.6689]	0.1200 [−0.0875, 0.3275]	**+0.3652; z = 2.58, p = 0.0098**
age	−0.5505 [−0.6081, −0.4930]	−0.5945 [−0.6752, −0.5139]	−0.4780 [−0.5717, −0.3844]	−0.1165; z = −1.85, p = 0.065
c_charge_degree_M	−0.1732 [−0.2797, −0.0667]	−0.1413 [−0.2919, 0.0092]	−0.1552 [−0.3355, 0.0251]	+0.0139; z = 0.12, p = 0.908

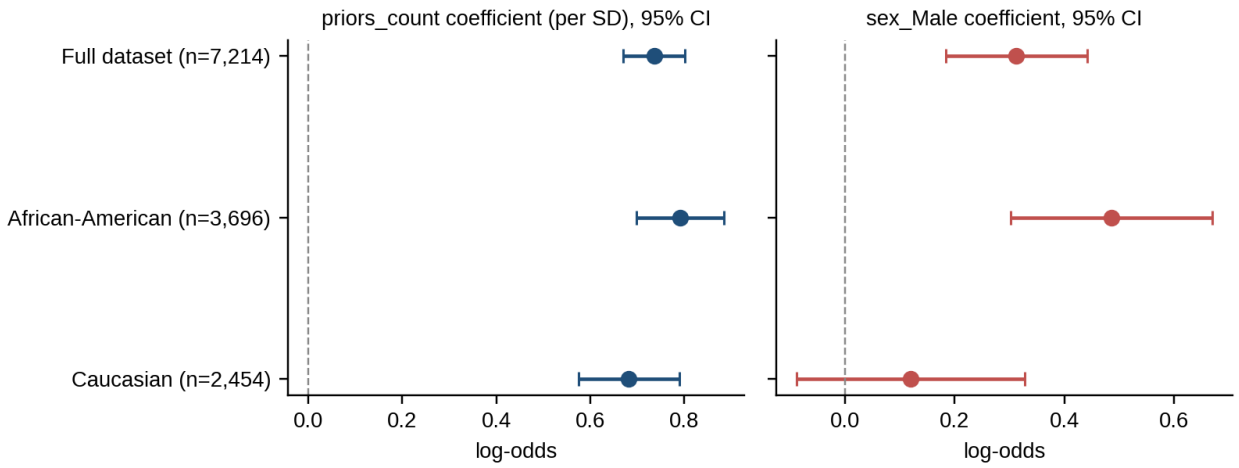


Figure 2. priors_count and sex_Male coefficients with 95% confidence intervals (dashed line = no effect).

Conclusion: no statistically significant reversal exists. The significantly different effect is **sex_Male**: 0.4852 (95% CI 0.3014 to 0.6689) for African-American defendants against 0.1200 (−0.0875 to 0.3275, p = 0.257) for Caucasian defendants. The difference of 0.3652 (95% CI 0.088 to 0.642; z = 2.58, p = 0.0098) still passes a Bonferroni threshold of 0.0125 across the four shared predictors. Because sex_Male is a 0/1 dummy, this comparison is not thrown off by the group-specific standardisation that affects the continuous coefficients. The age effect is also a bit stronger for African-American defendants (p = 0.065), but that is only a hint.

Explanation and policy implication are in Part 3, Question 2.

• 3.2 Omitted-variable bias

Dropped variable: race. Full model: logistic regression of two_year_recid on age, priors_count, sex, race and c_charge_degree (n = 7,214). Reduced model: the same without race (five dummy columns removed). Both models converged.

Table 5. Coefficients with and without race. % change = 100 × (reduced − full) / |full|.

Predictor	Full coefficient	Reduced coefficient	Absolute change	% change (signed)	|% change|
age	−0.5505	−0.5576	0.0071	−1.28%	1.28%
priors_count	0.7357	0.7520	0.0163	+2.22%	2.22%
sex_Male	0.3126	0.3032	0.0094	−3.01%	**3.01%**
c_charge_degree_M	−0.1732	−0.1776	0.0044	−2.55%	2.55%
intercept (not a predictor)	−0.3515	−0.3916	0.0401	−11.41%	11.41%

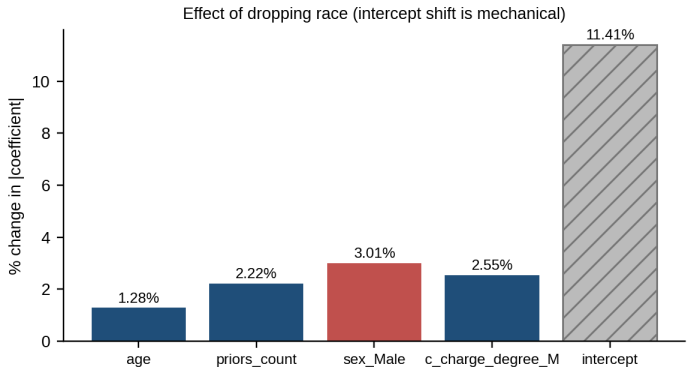


Figure 3. Percentage shift in each coefficient after dropping race. The intercept moves mainly because the baseline category changes.

Among the four predictors **sex_Male shifted the most in percentage terms (3.01%)**, followed by c_charge_degree_M (2.55%), priors_count (2.22%) and age (1.28%); priors_count has the largest absolute change (0.0163). No coefficient changed sign and every p-value stayed below 0.002 (charge degree: 0.0014 to 0.0011). In the full model the race terms are race_Caucasian −0.0399 (p = 0.493), race_Hispanic −0.2445 (p = 0.010), race_Other −0.2577 (p = 0.031), race_Asian −0.4782 (p = 0.247) and race_Native American 0.0422 (p = 0.935, n = 18). Discussion in Part 3, Question 3; detection of hidden confounds in Question 4.

• 3.3 Adversarial subset

On the 2,165 test cases I searched for P(reoffend) ≥ 0.75 with actual outcome 0 (false positive) or P ≤ 0.25 with actual outcome 1 (false negative). **115 cases (5.31% of the test set)** qualify, so I sampled 50 at random (random_state = 11). Among all 115, 36 are false positives (31.3%) and 79 false negatives (68.7%) (derived); among the 50 sampled, 12 are false positives (24%) and 38 false negatives (76%).

Table 6. The 50 sampled high-confidence errors compared with the full test set (n = 2,165).

Characteristic	Adversarial subset (n = 50)	Full test set (n = 2,165)
Mean age	49.48	34.90
Median age	50	31
25th / 75th percentile of age	44.5 / 54	25 / 43
Maximum age	96	96
Share aged 40 or older	80.0%	30.3%
Mean / median priors_count	4.00 / 1	3.44 / 2
Share with 3 or fewer priors	76.0%	69.5%

Characteristic	Adversarial subset (n = 50)	Full test set (n = 2,165)
Share with 9 or more priors	24.0%	11.9%
Male / Female	78.0% / 22.0%	80.83% / 19.17%
African-American / Caucasian	40.0% / 44.0%	51.18% / 32.24%
Hispanic / Other	10.0% / 6.0%	9.93% / 6.14%
Felony / Misdemeanour charge	54.0% / 46.0%	64.5% / 35.5%

Shares of age 40+, priors ≤ 3 and ≥ 9, and the full-test priors and charge figures are derived from the saved CSV files; the remaining entries are printed by the notebook.

Table 7. The 50 rows split by error type (derived from part2_3_adversarial_subset.csv).

Error type	n	Mean age (range)	Mean priors (range)	Mean predicted P
False negative (P ≤ 0.25, reoffended)	38	52.47 (37 to 96)	0.76 (0 to 3)	0.18
False positive (P ≥ 0.75, did not reoffend)	12	40.00 (24 to 64)	14.25 (9 to 20)	0.80

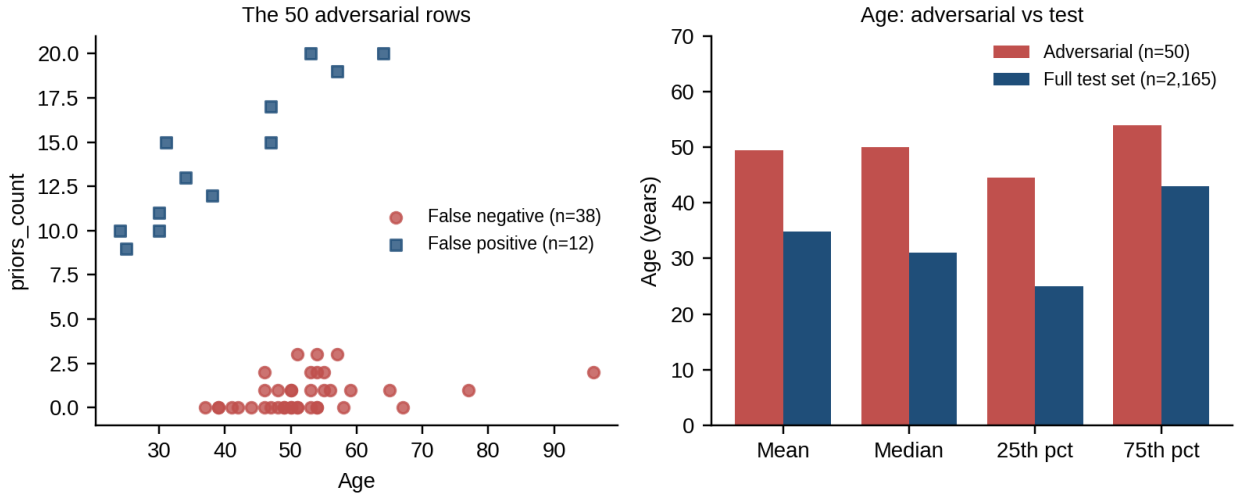


Figure 4. Left: the 50 adversarial rows by age and priors_count. Right: age of the subset versus the full test set.

Shared characteristic: age and prior record that push the score to an extreme. The subset is 14.6 years older than the test set on average, and 80% of its rows are aged 40 or more. 76% (38 rows) are older defendants (mean age 52.5) with 3 or fewer priors (mean 0.76) who reoffended, while 24% (12 rows) have 9 to 20 priors and did not reoffend. The mean of 4.00 priors hides this split (median 1). Caucasian defendants are over-represented (44.0% versus 32.24%), making up 20 of the 38 false negatives, while African-American defendants make up 10 of the 12 false positives; with n = 50 these counts are only indicative. The statistical explanation is in Part 3, Question 5.

• 3.4 Fairness trade-off analysis

Sensitive variable: race. Group A = African-American (1,108 test defendants) and Group B = Caucasian (698). The logistic probability threshold was swept from 0.05 to 0.95 in steps of 0.10. DP = $P(\hat{Y}=1|A)/P(\hat{Y}=1|B)$; EO = $\min(TPR_A/TPR_B, FPR_A/FPR_B)$; PP = PPV_A/PPV_B. Because a flag means 'high risk', a ratio above 1 means Group A is flagged or wrongly flagged more often (for PP, that Group A's flags are more often correct). Accuracy is computed on all 2,165 test defendants.

Table 8. Fairness metrics across the threshold sweep (highlighted row = the threshold I recommend). Ideal value for DP, EO and PP is 1.0.

Threshold	DP	EO	PP	Accuracy
0.05	1.0000	1.0000	1.3479	0.4503
0.15	1.0708	1.0285	1.2947	0.4855
0.25	1.2686	1.1370	1.2082	0.5483

Threshold	DP	EO	PP	Accuracy
0.35	1.4237	1.2840	1.2157	0.6115
0.45	1.7705	1.6264	1.2764	0.6517
0.55	2.2809	1.7252	1.0195	0.6577
0.65	2.4764	1.8585	1.0116	0.6189
0.75	3.0373	2.5237	1.1200	0.5963
0.85	4.4097	3.2047	0.9796	0.5737
0.95	3.7798	2.5704	0.9167	0.5552

Table 9. Group-level rates behind Table 8 (A = African-American, B = Caucasian), from part2_4_fairness_thresholds.csv.

t	Flag rate A	Flag rate B	TPR A	TPR B	FPR A	FPR B	PPV A	PPV B
0.05	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	52.5%	39.0%
0.15	98.6%	92.1%	99.8%	97.1%	97.3%	89.0%	53.2%	41.1%
0.25	92.7%	73.1%	96.6%	84.9%	88.4%	65.5%	54.7%	45.3%
0.35	82.4%	57.9%	91.6%	71.3%	72.2%	49.3%	58.4%	48.0%
0.45	62.9%	35.5%	77.7%	46.3%	46.6%	28.6%	64.8%	50.8%
0.55	37.9%	16.6%	51.4%	29.8%	23.0%	8.2%	71.2%	69.8%
0.65	20.6%	8.3%	29.4%	15.8%	10.8%	3.5%	75.0%	74.1%
0.75	12.2%	4.0%	18.6%	7.4%	5.1%	1.9%	80.0%	71.4%
0.85	5.1%	1.1%	8.2%	2.6%	1.5%	0.2%	85.7%	87.5%
0.95	1.1%	0.3%	1.9%	0.7%	0.2%	0.0%	91.7%	100.0%

At t = 0.05 almost everyone is flagged, so DP = EO = 1.0 is trivial (accuracy 0.4503 is essentially the 45.08% base rate) and the PP of 1.3479 simply reflects the group base rates (52.5% versus 39.0%). At t = 0.95 only about 12 African-American and 2 Caucasian defendants are flagged (derived), so the EO rebound and the PP of 0.9167 are mostly noise from tiny samples.

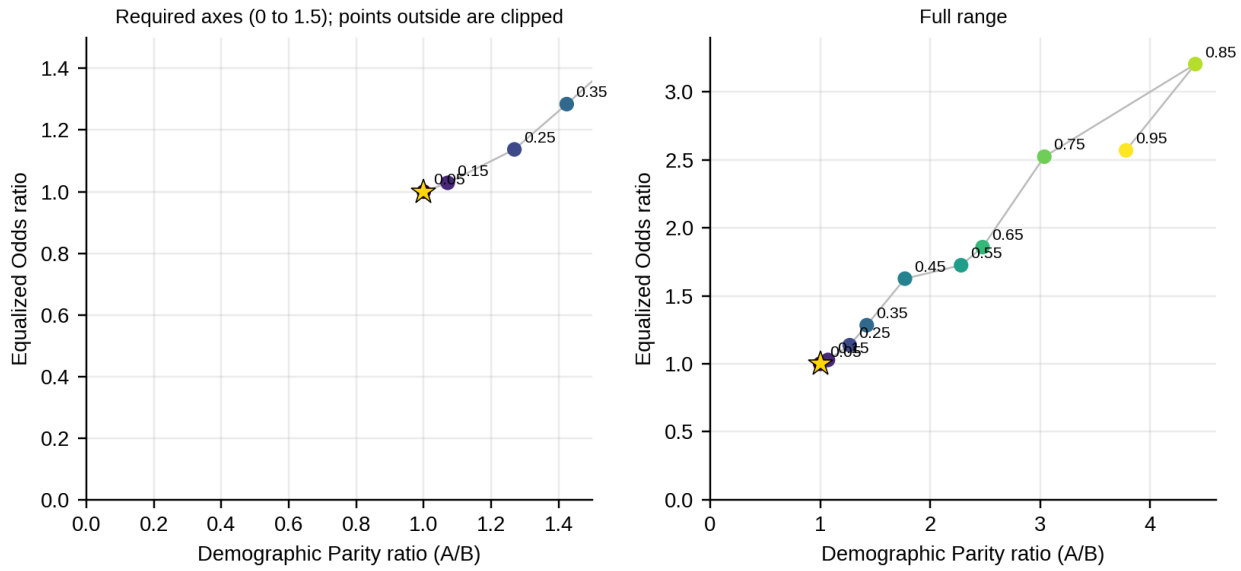


Figure 5. Trade-off chart: DP (x) against EO (y), each point labelled with its threshold; the star marks the ideal (1, 1). The left panel uses the required 0–1.5 axes, so every point from t = 0.45 onward lies off-chart; the right panel shows the full range.

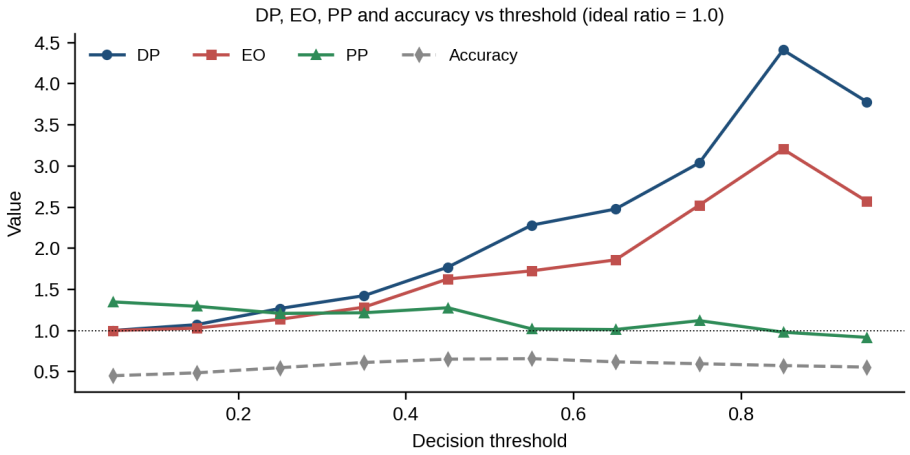


Figure 6. DP, EO, PP and accuracy as functions of the threshold.

Quantifying the trade-off (percentages are changes in the ratio; the cost ratios are simple arithmetic on Table 8, derived). (a) Lowering t from 0.55 to 0.35 improves DP from 2.2809 to 1.4237 (a 37.6% gain) and EO from 1.7252 to 1.2840 (a 25.6% gain), but PP moves from 1.0195 to 1.2157, which is 19.2% further from the ideal, and accuracy falls from 0.6577 to 0.6115 (−4.62 points). The trade-off cost is 19.25 / 37.58 = **0.51** percent of PP lost per percent of DP gained (0.75 per percent of EO). (b) Conversely, raising t from 0.35 to 0.65 improves PP by 16.8% (1.2157 to 1.0116) while DP worsens by 73.9% (to 2.4764) and EO by 44.7% (to 1.8585); the cost is 73.95 / 16.79 = **4.40** percent of DP per percent of PP (2.66 for EO), with accuracy up only 0.74 points.

Key insight: DP and EO move together here, while PP moves the other way, so no threshold satisfies all three. PP is closest to 1 (1.0116 at t = 0.65, 1.0195 at t = 0.55) exactly where DP (2.4764, 2.2809) and EO (1.8585, 1.7252) are worst. The definition of EO as a minimum also hides the larger disparity: at t = 0.55 the false positive ratio is 2.80 although EO reports 1.7252 (the true positive ratio). I recommend prioritising equalized odds and using t = 0.35 (justified in Part 3, Questions 7 and 8).

4. Part 3: Descriptive analysis questions

• Question 1

Explain in detail why the threshold-based approach and the direct logistic regression approach did, or did not, flag the same individuals as "high risk" in your data. Quote the specific count/percentage of mismatched cases. Reference the median predicted value from linear regression and how it compares to the 0.5 threshold in logistic regression.

– The two rules flagged different people in 323 of the 2,165 test cases (14.92%) and agreed on the other 1,842 (85.08%). The confusion matrix, with rows for the linear median rule and columns for the logistic 0.5 rule, is [[1066, 16], [307, 776]]. What stands out is how one-sided it is: 307 defendants (95.0% of the mismatches) are high risk under the linear rule but low risk under logistic regression, and only 16 go the other way. I think the cut-off explains almost all of this, not the models themselves. The median of Model A's test predictions is 4.6267 on the 1–10 decile scale. Squeezed onto a 0–1 scale as (4.6267 − 1)/9, that is about 0.40, clearly lower than the 0.5 that logistic regression uses. A median rule has to flag exactly half the people, so Model A flags 1,083 (50.02%), while Model B flags 792 (36.58%). In reality only 45.08% of test defendants (976 of 2,165) reoffended. The 307 extra flags are borderline cases: their logistic probabilities average 0.447 and run from 0.326 to 0.499, while the 16 reverse cases sit just above 0.5 (mean 0.526). The two models actually rank people almost the same way (Spearman correlation 0.95), so a gap of roughly 0.1 between the cut-offs produces most of the 323 mismatches. There are two smaller reasons. Model A is copying COMPAS's own decile_score rather than actual recidivism, and OLS is not bounded, so its predictions run from −2.47 to 12.21, outside the 1–10 scale. Performance moves with the cut-off too: Model B at 0.5 has accuracy 0.6628, precision 0.6553 and recall 0.5318, while Model A's median rule has 0.6420, 0.5928 and 0.6578. So neither rule is neutral. Each one is a decision about how many people to label high risk.

• Question 2

Describe the subgroup you found (or the subgroup with significantly different effects). Using your actual regression coefficients from the full model, subgroup A, and subgroup B, explain why the trend reverses or differs. Cite the exact coefficient values and confidence intervals. Explain what policy conclusion would change if the subgroup were ignored.

– I split the data by race and compared African-American defendants (3,696) with Caucasian defendants (2,454), alongside the full data (7,214). For the key predictor, priors_count, nothing flips sign. The full model gives 0.7357 (95% CI 0.6698 to 0.8016), African-American defendants 0.7905 (0.6973 to 0.8836) and Caucasian defendants 0.6816 (0.5743 to 0.7889). The odds ratio per standard deviation is 2.20 against 1.98, but the intervals overlap, and a Wald test on the 0.1089 gap gives z = 1.50, p = 0.133, so I can't call that difference real. The effect that does differ significantly is sex. The sex_Male coefficient is 0.4852 (0.3014 to 0.6689) for African-American defendants but just 0.1200 (−0.0875 to 0.3275, p = 0.257) for Caucasian defendants, while the pooled model says 0.3126 (0.1836 to 0.4416). The gap of 0.3652 (95% CI 0.088 to 0.642) gives z = 2.58, p = 0.0098, and it still holds after a Bonferroni correction across the four shared predictors (threshold 0.0125). In odds terms, being male multiplies the odds of recidivism by 1.62 for African-American defendants, 1.13 for Caucasian defendants and 1.37 in the pooled model, so the pooled figure is an average that fits neither group. My guess, which this data cannot test, is that men in the African-American group carry extra risk that the model pins on sex, maybe through things we do not measure, such as how heavily they are policed. If someone ignored the subgroups, they would read the pooled 0.3126 as sex raising risk by about 37% for everybody, and that could be used to justify sex-based supervision rules that mostly land on African-American men. The safer conclusion is that the model behaves differently across racial groups, so each group needs its own audit.

• Question 3

State which variable you dropped for the omitted-variable test. Report the specific before/after shift in coefficients for at least 4 predictors—include a table. Calculate the percentage change for each. Which coefficient shifted the most? What does this reveal about how much you can trust a model you did not stress-test this way?

Coefficients before and after dropping race (same values as Table 5).

Predictor	Full	Reduced (no race)	% change
age	−0.5505	−0.5576	−1.28%
priors_count	0.7357	0.7520	+2.22%
sex_Male	0.3126	0.3032	−3.01%
c_charge_degree_M	−0.1732	−0.1776	−2.55%

– I dropped race (six categories, so five dummy variables, with African-American as the baseline). My reasoning was that race could be tied to age and arrest history through patterns of policing, and also to recidivism itself. Both models are logistic regressions on all 7,214 defendants with the numeric predictors standardized. The full model uses age, priors_count, sex, race and charge degree, and the reduced one leaves out race. The table above shows the results, with percentage change defined as 100 × (reduced − full)/|full|. Among the four predictors, sex_Male shifted the most in percentage terms at −3.01% (0.3126 to 0.3032). Next come c_charge_degree_M at −2.55%, priors_count at +2.22% (0.7357 to 0.7520, the biggest absolute change at 0.0163) and age at −1.28%. The intercept moved 11.41% (−0.3515 to −0.3916), but that is mostly mechanical, because dropping race changes the baseline from African-American defendants to a mix of groups. No sign flipped, and every predictor stayed significant (p < 0.002). So for this particular confound the model looks stable: race is not tangled up with the other four coefficients. That is reassuring, but it only shows that these four numbers do not lean much on race. It does not show the model is free of confounds, and without running the test I would have had no evidence either way. There is also a catch: priors_count may itself carry the effect of unequal policing, and if so, controlling for it could hide a race effect that the full model only weakly shows (race_Caucasian −0.0399, p = 0.493). I would trust a model that has not been stress-tested like this for ranking people (ROC-AUC 0.7095) much more than for explaining anything or setting policy from its coefficients. Small coefficients, like −0.1732 for charge degree, are especially fragile.

• Question 4

As a practitioner, how would you detect a hidden confound in your model that you cannot currently see? Propose one concrete, real-world method (e.g., domain expert audit, external validation on a different dataset, residual analysis, instrumental variable approach). This question has no clean answer; you will be graded on the realism and concreteness of your proposal.

– A hidden confound affects both the inputs and the outcome but is missing from my data, so my coefficients can't show it. For this COMPAS model I would run a three-step audit. Step 1 is an expert review. I would show five to eight probation officers, public defenders and pretrial-services staff the 50 adversarial rows (38 false negatives with mean age 52.5 and at most 3 priors, and 12 false positives with 9 to 20 priors) and ask what information outside the form would explain each surprise, such as employment, housing stability, substance treatment or where the arrest took place. Step 2 is to turn their ideas into measurable fields. The file already has some unused ones, like juv_fel_count, juv_misd_count and c_charge_desc, and I would collect the rest for a fresh sample of about 1,000 defendants from a later year or another county. Step 3 is the test. I would add each candidate to the model and see whether existing coefficients move past a margin agreed in advance, say 10% (my race test moved them by only 1.3% to 3.0%), or whether a likelihood-ratio test gives $p < 0.05$, and I would look at residuals by subgroup. I would also run the unchanged model on the fresh sample. If ROC-AUC falls well below 0.7095 or accuracy below 0.6628 (a drop of more than about 0.05 would trigger a review), something unmeasured or a distribution shift is probably at work. My biggest suspicion is the outcome itself. two_year_recid records a re-arrest, which depends on how heavily an area is policed, so I would also compare against another outcome such as a conviction for a new offence. It is affordable, but it cannot prove that no confound is left; it only lowers the chance that I am blind to a big one.

• Question 5

Describe your 50-row adversarial subset. What common characteristic do these rows share, and what percentage of the subset exhibits this trait? Statistically, why does this characteristic cause your model to fail confidently? Reference your model's coefficients or learned decision boundaries to explain the failure mechanism.

– The test set has 115 high-confidence errors (5.31% of 2,165), and I randomly sampled 50 of them (random_state = 11). Of those 50, 38 (76%) are false negatives ($P \leq 0.25$, mean 0.18, but the person did reoffend) and 12 (24%) are false positives ($P \geq 0.75$, mean 0.80, but the person did not). What they share is that age and prior record push the model's score to an extreme: 76% are older defendants (mean age 52.5) with 3 or fewer priors, and 24% have 9 to 20 priors. The subset as a whole is much older than the test set (mean age 49.48 versus 34.90, and 80% are 40 or older versus 30.3%), and its mean of 4.00 priors hides a split (median 1). Caucasian defendants are 44.0% of the subset against 32.24% of the test set. They make up 20 of the 38 false negatives, while African-American defendants make up 10 of the 12 false positives, though the counts are small. The cause is the model's additive form, with one straight-line slope per variable: −0.5505 per SD of age and +0.7357 per SD of priors_count (age SD 11.89, priors SD 4.88). A 52-year-old with 0 or 1 priors gets about −0.8 from age and −0.4 to −0.5 from priors, which gives P near 0.18, yet roughly one in five such people still reoffends. A defendant with 14 priors, 2.2 SD above the mean, gets about +1.6, giving P near 0.80. Interestingly, the model is not overconfident. $P \geq 0.75$ predictions are wrong 20.7% of the time (36 of 174) and $P \leq 0.25$ predictions 21.4% (79 of 369). It fails because it has no age × priors interaction and ignores other fields, so it cannot pick out the fifth who break the pattern.

• Question 6

Using your own fairness trade-off chart from Part 2.4, explain in concrete terms what the trade-off cost means for a real person affected by this decision. Describe a specific, realistic scenario: two individuals (one from each demographic group) and their outcomes under different decision thresholds. Show how prioritizing one fairness metric harms another group.

– Picture two defendants with identical records: Marcus (African-American, Group A) and Daniel (Caucasian, Group B). Both are 27, male, charged with one felony, have two priors, and neither will reoffend. The test set has 1,108 African-American and 698 Caucasian defendants. At t = 0.55 the model is at its most accurate (0.6577) and nearly satisfies predictive parity (PP = 1.0195, as flagged defendants reoffend 71.2% versus 69.8% of the time). But DP = 2.2809 and EO = 1.7252. In practice, 37.9% of African-American defendants get flagged against 16.6% of Caucasian defendants, and among people who do not reoffend the false positive rates are 23.0% and 8.2%. So Marcus is about 2.8 times as likely as Daniel to be wrongly labelled high risk. Now lower the threshold to 0.35 to even things out. DP falls to 1.4237, EO to 1.2840 and the false positive ratio to 1.47. But Marcus's chance of a wrongful flag jumps from 23.0% to 72.2%, and Daniel's from 8.2% to 49.3%, a six-fold increase. The price shows up in PP = 1.2157: a flag on Daniel is right only 48.0% of the time, against 58.4% for Marcus's group, so Daniel's label is wrong more often. Accuracy also drops 4.62 points to 0.6115. Going from 0.55 to 0.35 improves DP by 37.6% and EO by 25.6% but worsens PP by 19.2%, a cost of 0.51. If we protect predictive parity, the label stays equally believable, but the false flags pile up on Marcus's group. If we push DP or EO, the false flags spread to both groups and the label becomes less reliable for Daniel's. Either way, a real person pays.

• Question 7

Choose a specific decision threshold from your sweep (e.g., t=0.35). Argue, using at least 3 specific numbers from your analysis (DP, EO, PP, accuracy, subgroup-specific rates), why this threshold is defensible from a fairness standpoint. Describe the strongest counter-argument a critic could raise against your choice and acknowledge its validity.

– I picked t = 0.35. At this threshold DP = 1.4237, EO = 1.2840 and PP = 1.2157, so the biggest distance from the ideal 1.0 is 0.42. That is the smallest of any threshold whose accuracy beats the best trivial classifier: accuracy here is 0.6115, against 0.5492 for flagging nobody and 0.4508 for flagging everybody. Lower thresholds are not usable. At t = 0.25 the biggest gap is smaller (0.27), but accuracy is 0.5483, below the flag-nobody baseline, and 92.7% of African-American defendants get flagged. Compared with t = 0.55 (biggest gap 1.28, accuracy 0.6577), t = 0.35 shrinks the DP gap by 66.9% and the EO gap by 60.8% for a loss of 4.62 accuracy points, and the false positive ratio drops from 2.80 to 1.47 (72.2% versus 49.3%). The strongest counter-argument is levelling down. The ratios look better mostly because 82.4% of African-American and 57.9% of Caucasian defendants are now flagged high risk, so the real harm to people who never reoffend grows in both groups. The false positive rate for African-American defendants is 72.2%, up from 23.0% at t = 0.55. A critic could add that EO, defined with min(), hides the worse disparity (the false positive ratio of 1.47 is bigger than the true positive ratio of 1.28), that PP = 1.2157 means flags on Caucasian defendants are wrong more often, and that accuracy of 0.6115 is 5.1 points below the 0.5-threshold model (0.6628). I think all of that is valid. That is why I see 0.35 as the least-bad fairness compromise for analysis, and not as a threshold anyone should use to justify detention.

• Question 8

If a policymaker had to choose between prioritizing demographic parity or equalized odds for your dataset's specific context, which would you recommend, and why? Cite your own trade-off numbers from Part 2.4. Acknowledge what fairness metric you are sacrificing and why that is acceptable in your view.

– For recidivism prediction I would put equalized odds ahead of demographic parity. The decision depends on an outcome, so what matters is that people who will not reoffend are wrongly labelled high risk at similar rates whatever their race, and that people who will reoffend are caught equally. Demographic parity ignores the outcome, and the base rates in my data differ: 52.5% of African-American test defendants reoffend versus 39.0% of Caucasian ones (the PPV at t = 0.05, where nearly everyone is flagged). Forcing equal flag rates would mean labelling more low-risk Caucasian defendants or fewer high-risk African-American ones. In my sweep, though, the two metrics move together, so choosing EO costs little DP: from t = 0.55 to t = 0.35, EO improves from 1.7252 to 1.2840 and DP from 2.2809 to 1.4237 (a 37.6% gain). The metric I give up is predictive parity. PP rises from 1.0195 to 1.2157, a 19.2% departure from 1.0, which means flagged Caucasian defendants are wrong more often than flagged African-American defendants (PPV 48.0% versus 58.4%). The cost works out to 0.51 percent of PP per percent of DP gained, or 0.75 per percent of EO. I can live with that because a false positive here can mean detention or tougher supervision, and because PP is the easiest metric to satisfy just by raising the threshold, which can hide a false positive ratio of 2.80 (at t = 0.55). I have two cautions. The min() in the EO definition hides the larger false positive gap (1.47 against 1.28 for true positives at t = 0.35), and the base-rate gap may itself come from unequal policing, in which case DP should count for more. That is a values judgement, and the data cannot settle it.

5. Part 4: Reflection

“If a policy-maker asked you: ‘just give me one number: should we deploy this model or not?’, what would you say, and what is the limitation of that ‘one number’ answer that you discovered yourself through your analysis?”

Direct answer. My answer would be no, not as it stands. I would reject it as an automated decision-maker, and consider advisory use only with human review once the fairness gaps are dealt with. The headline reason is that accuracy is just 0.6628 and no threshold gives African-American and Caucasian defendants similar error rates.

Why a single number fails. First, accuracy hides who is wrong. The model's 0.6628 is only 11.4 points better than the 0.5492 you get by flagging nobody, it catches just 53.18% of people who actually reoffend, and it averages over groups whose base rates differ (52.5% of African-American versus 39.0% of Caucasian test defendants reoffend). Second, the threshold is a values question, not a fact in the data. Going from t = 0.55 to 0.35 costs 4.62 accuracy points but shrinks the DP gap by 66.9%. Whether that is worth it depends on how a society weighs a wrongful flag against a missed reoffender, and no dataset can answer that. Even choosing between my two models is a threshold choice, since the median rule and the 0.5 rule disagree on 323 people. Third, fairness metrics conflict. Predictive parity is nearly perfect at t = 0.55 (1.0195), exactly where DP (2.2809) and EO (1.7252) are poor and the false positive rates are 23.0% versus 8.2%. A single "fairness number" would quietly pick a winner. Fourth, effects differ by subgroup. The sex_Male coefficient is 0.4852 for African-American and 0.1200 for Caucasian defendants (p = 0.0098), and the pooled 0.3126 hides that. Fifth, the model fails confidently on predictable subsets: 5.31% of test cases are high-confidence errors, and 76% of my 50-row sample are older defendants with few priors. Sixth, hidden confounds stay unknowable. Dropping race moved four coefficients by at most 3.01%, which only reassures me about that one variable, and the outcome itself (re-arrest) may reflect policing more than behaviour. Seventh, one number carries no uncertainty. Only 698 Caucasian defendants are in the test set, and at t = 0.95 about two of them are flagged, so ratios at the extremes are mostly noise. Squeezing all of this into one number gives a false sense of certainty, and the person who has to decide deserves to see the pieces that got squashed together. If I had to put it in a sentence, I would say that the model ranks people reasonably well (ROC-AUC 0.7095) but cannot be trusted to decide anything about them on its own.

Grounding in evidence. Everything above rests on my own results. 323 of 2,165 defendants (14.92%) change class depending on the cut-off (Part 1). Dropping race shifts coefficients by 1.28% to 3.01% (2.2). There are

115 confident errors, 5.31% of the test set (2.3). And DP, EO and PP are 2.2809, 1.7252 and 1.0195 at t = 0.55 but 1.4237, 1.2840 and 1.2157 at t = 0.35 (2.4).

Closing. The big lesson for me is that responsible deployment means being open about trade-offs instead of hiding them behind one metric. A release I could defend would document the chosen threshold, the known failure regions and the unresolved fairness gaps, would keep monitoring group-level error rates, and would give every flagged person a clear way to appeal.